

CUT: Chromatic-temporal U-Transformer for Expressive Multi-track MIDI Parameter Estimation

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Abstract—Expressive Performance Rendering (EPR) is a major challenge in computational music research, aiming to transform quantized, mechanical musical scores into performances with human-like nuance. While significant progress has been made in modeling expression for solo instruments, the task of estimating expressive parameters directly from deadpan, multi-track MIDI sequences remains a substantial challenge. This gap is primarily due to the lack of effective input representations and learning paradigms capable of handling the intricate interplay between multiple instruments. In this work, we formally define the task of Multi-track Expressive Parameter Estimation and propose a novel solution. We introduce *CUT*, a Chromatic-temporal U-Transformer that processes music via a multi-channel piano-roll representation using a self-attention mechanism across both temporal (frame-wise) and chromatic (semitone-wise) dimensions. To enable this research, we curated a high-quality dataset of 87,989 multi-track MIDI samples. Our model demonstrates superior parameter estimation performance, achieving a frame-level duration F1 score of 93.47%, a velocity Mean Absolute Error (MAE) of 6.97, and a Mean Opinion Score (MOS) that is only 0.14 points below the human-generated ground truth—outperforming the baseline by a significant margin of +0.69 MOS. These advancements hold substantial implications for enhancing the perceptual fidelity of multi-track music transcription and enabling the large-scale restoration of low-quality MIDI data.

Index Terms—Music Information Retrieval, Computer Generated Music, Expressive Performance Rendering

I. INTRODUCTION

Musical performance is characterized not only by the sequence of discrete notes but also by the nuanced variations in timing, dynamics, and articulation that convey musical expression. In the fields of computer music and artificial intelligence, the task of incorporating these performance elements into mechanical, “deadpan” representations (e.g., musical scores or quantized MIDI) is formally known as *Expressive Performance Rendering (EPR)* [1].

Recent years have witnessed promising advancements in EPR, exploring various input-output paradigms—ranging from Score-to-MIDI [2] and MIDI-to-Audio [3] to Audio-to-Audio [4], [5] translation—and diverse control protocols (text-conditioned [6], emotion-conditioned [7], unconditioned [8], etc.). Despite this progress, the field currently lacks a systematic, unified definition, leading to a scarcity of universal paradigms and standardized evaluation metrics. Furthermore, existing research predominantly focuses on monophonic instruments or single-track polyphony (most notably the piano) [9], leaving the complex domain of multi-track polyphonic music largely unexplored.

To address these limitations, we propose a hierarchical formalization of the EPR task, dividing it into two distinct stages.

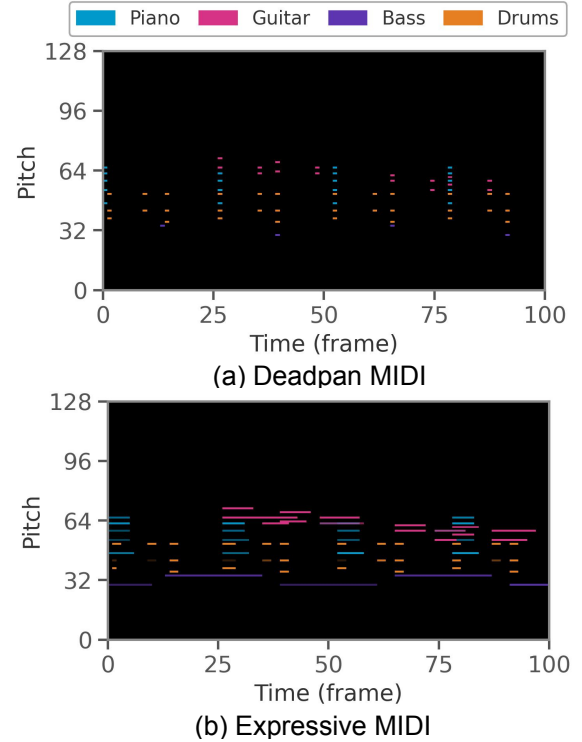


Fig. 1. Visual comparison between (a) deadpan (quantized) MIDI and (b) expressive performance MIDI across four tracks: piano, guitar, drum, and bass. The horizontal length of each note represents its duration (articulation), while the color intensity indicates the velocity (dynamics).

The first stage, *Parameter Estimation*, focuses on generating expressive parameters for symbolic representations (MIDI) from quantized inputs, which converts deadpan MIDI to expressive MIDI as shown in Fig.1. The second stage, *Nuance Rendering*, involves the synthesis of audio that incorporates micro-timing deviations, instrument-specific techniques, and timbral details.

In this work, we focus on the first stage, extending the scope of Parameter Estimation to the multi-track polyphonic domain. Our objective is to estimate performance-critical attributes—specifically *velocity* and *duration*—based on only quantized onset information. Velocity introduces dynamic variation and “breathing” qualities to the music, while duration dictates articulation, playing a decisive role in the rhythmic perception of the piece. This task holds significant value for real-world applications, such as serving as a creative assistant in Digital Audio Workstations (DAWs), restoring low-quality web MIDI files, and acting as a post-processing stage for multi-track Automatic Music Transcription (AMT) [10] to enhance the listenability of transcribed results.

In this work, we present a novel, effective end-to-end training paradigm and a high-performance generative model for multi-track MIDI parameter estimation. We target four representative instrument tracks: piano, guitar, bass, and drums, utilizing a multi-channel image-like piano-roll representation. To efficiently process this high-dimensional data, we introduce a Chromatic-Temporal (C-T) patching technique, which enables the model to capture both semitone-wise and frame-wise dependencies. This approach significantly reduces the input token sequence length while maintaining the high temporal resolution required for precise point-to-point generation. For the backbone architecture, we employ a U-shaped Transformer with convolutional skip connections, facilitating the simultaneous modeling of long-range musical structures and the extraction of multi-scale local features. Furthermore, to address the sparsity and boundary sensitivity inherent in piano-roll estimation targets, we design a position-specific loss weighting scheme named *Region-Adaptive Weighted (RAW) Loss*, ensuring the predictive consistency of sparse regions and the positional accuracy of boundaries for expressive parameters.

We constructed a dataset comprising 87,989 high-quality samples filtered from large-scale public multi-track MIDI databases. Experimental results demonstrate that our model achieves state-of-the-art performance, recording a note-level Offset F1 score of 86.09% and a Velocity F1 score of 76.25% across the four instrument tracks. In subjective listening tests, our model outperformed the baseline by 0.69 points in MOS for overall quality and only lagged behind the ground truth by 0.14 points. These results underscore the immense potential of our model in practical scenarios and provide a solid foundation for future research in performance rendering.

II. RELATED WORK

As previously discussed, EPR encompasses a diverse range of input-output task configurations. Based on the granularity and density of the input information, we categorize existing approaches into two distinct paradigms: *high-informed* and *low-informed* EPR.

A. High-Informed EPR

Research on high-informed EPR predominantly focuses on learning the mapping from information-rich symbolic representations—such as MusicXML or full musical scores—to expressive performances. These inputs typically contain explicit musical metadata, including nominal note durations, tempo specifications, bar lines, time signatures, and articulation markings (e.g., legato and staccato). Prominent works in this domain include VirtuosoNet [11] and RenderBox [6]. A representative state-of-the-art framework is MIDI-DDSP [3], which introduces a hierarchical note-performance-synthesis architecture. MIDI-DDSP [5] converts MIDI sequences into high-fidelity audio by predicting expressive performance parameters (e.g., vibrato, dynamics, and brightness) and leveraging Differentiable Digital Signal Processing (DDSP) to achieve fine-grained control over musical timbre and expression.

However, a fundamental limitation of high-informed models lies in their reliance on explicit priors. Since the complete score information is explicitly provided, these models often struggle to capture the latent structural information and the intrinsic correlations between musical elements, as the architecture is not compelled to infer these dependencies from the data itself.

B. Low-Informed EPR

Low-informed EPR aligns more closely with the problem setting of this study, where large-scale Transformer-based models are employed to learn musical representations under weakly supervised conditions. He et al. proposed Peransformer [12], a low-informed expressive performance rendering system that integrates a score-aware discriminator within an adversarial training setup. This approach enhances the realism and structural fidelity of the predicted performance parameters directly from raw MIDI inputs. Similarly, You et al. developed the Pianist Transformer [13], which pioneers a scalable self-supervised pre-training paradigm using large-scale unpaired MIDI data. By modeling nuanced expressive performance features without paired supervision, Pianist Transformer significantly outperforms models trained solely on limited paired datasets.

Our proposed task strictly falls under the low-informed category. The model accepts only minimal information—specifically note onsets and pitch as input, without requiring additional annotations beyond high-quality MIDI files. Research on such low-informed joint estimation of multi-track velocity and duration parameters remains scarce; existing works only tackled single-track estimation issue [9], [14]. Furthermore, our framework can serve as a pre-processing module for high-informed EPR systems, effectively bridging the gap between deadpan MIDI sequences and fully expressive audio generation.

III. METHODS

A. Input Representation

To formally represent the musical input, we adopt a multi-track piano-roll representation denoted as a tensor $\mathcal{X} \in \mathbb{R}^{N \times C \times T}$. Here, $N = 4$ corresponds to the instrument tracks, specifically mapped to Piano, Guitar, Bass, and Drums (tracks 0, 1, 2, and 3, respectively). The dimension $C = 128$ represents the full MIDI pitch range, while T denotes the temporal dimension. To balance auditory fidelity with computational efficiency, we establish a time resolution of 20ms per frame. We fix the input sequence length at 60 seconds, resulting in a total temporal dimension of $T = 3000$ frames. An individual element $x_{n,c,t} = v$ signifies a note event with normalized velocity v occurring for instrument n at pitch c and time step t . To define the task as learning the mapping from plain onsets to expressive performance, we apply a strict preprocessing filter to the input: we retain only the onset frames for all note events and normalize their velocity values to a fixed scalar $v = 0.5$ (corresponding to the MIDI default of 64). Consequently, the input tensor $\mathcal{X}_{in}$ contains only binary onset

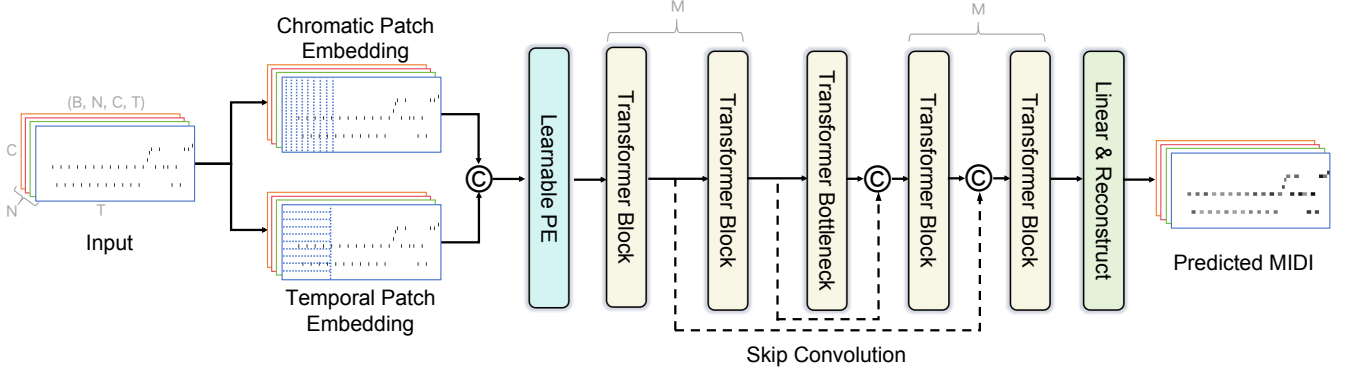


Fig. 2. The overall architecture of our proposed Chromatic-temporal U-Transformer (CUT) model for multi-track MIDI parameter estimation.

information, forcing the model to infer dynamics and duration solely from structural context.

To effectively capture both the instantaneous harmonic alignment across instruments and the long-term spectral evolution within specific frequency bands, we introduce a dual-granularity patching strategy termed *Chromatic-Temporal Patching*. Unlike standard patching methods that utilize uniform grids, we decompose the input tensor into two distinct patch types designed to impose specific inductive biases. First, we extract *Chromatic Patches* of shape $(4, 128, 5)$, which encapsulate the full pitch range over a short temporal window; this allows the model to explicitly attend to vertical dependencies such as chord voicings and inter-instrument synchronization. Second, we extract *Temporal Patches* of shape $(4, 16, 200)$, which capture long-range melodic contours and rhythmic patterns within narrower pitch bands. Both patch types are flattened and projected into a shared latent space via a linear transformation to a dimension of $D = 512$. The resulting embedding sequences from both views are concatenated, forming a unified sequence representation. Finally, to preserve the structural order of the original musical signal, we add learnable positional embeddings to this sequence before feeding it into the backbone model.

B. Backbone Architecture

As illustrated in Fig. 2, we employ a symmetric, constant-resolution U-shaped Transformer as the backbone model [15]. The architecture is composed of a contracting encoding path and an expansive decoding path, coupled via enhanced long-range skip connections. The encoding path consists of M stacked standard Transformer encoder blocks [16], which progressively extract hierarchical latent representations from the input sequence. Correspondingly, the decoding path comprises M Transformer blocks. Unlike standard U-Net architectures that directly concatenate features, we introduce a *Skip Convolution* module to mitigate the semantic gap between the encoder and decoder features and to inject local inductive biases [17]. Specifically, before fusion, the feature tokens from the encoding path are processed by a depthwise 1D convolution

with a kernel size of 3, followed by Layer Normalization. This operation aggregates local context along the sequence dimension without increasing parameter complexity. The data flow for the l -th decoding layer is formally defined as:

$$\begin{aligned} \mathbf{H}_{\text{enc}}^{(l)} &= \mathcal{T}_{\text{enc}}(\mathbf{H}_{\text{enc}}^{(l-1)}), \quad l \in [1, M] \\ \mathbf{Z}_{\text{skip}}^{(l)} &= \text{LN} \left(\text{Conv1d} \left(\mathbf{H}_{\text{enc}}^{(M-l+1)} \right) \right) \\ \mathbf{H}_{\text{dec}}^{(l)} &= \mathcal{T}_{\text{dec}} \left(\text{Linear} \left([\mathbf{H}_{\text{dec}}^{(l-1)}; \mathbf{Z}_{\text{skip}}^{(l)}] \right) \right) \end{aligned} \quad (1)$$

where $\mathbf{H}_{\text{enc}}^{(l)}$ and $\mathbf{H}_{\text{dec}}^{(l)}$ denote the output hidden states of the l -th layer in the encoding and decoding paths, respectively. $\mathbf{Z}_{\text{skip}}^{(l)}$ represents the locally enhanced skip features derived from the corresponding encoder layer. The Conv1d operator denotes a depthwise convolution (where the group size equals the channel dimension) applied along the sequence axis, effectively mixing information from adjacent tokens. The operator $[\cdot; \cdot]$ represents concatenation along the channel dimension. $\mathcal{T}$ refers to the standard Transformer block operations. A linear projection layer is applied post-concatenation to fuse the features and restore the channel dimensionality before passing the sequence to the subsequent attention mechanism.

Upon the completion of the decoding process, the final hidden states are passed through a prediction head. This module utilizes a linear projection to map the latent dimension back to the flattened patch size, followed by a reconstruction operation that reshapes the sequence into the original tensor dimensions (N, C, T) . This yields the predicted multi-channel piano roll, which retains the exact spatial and temporal resolution of the input.

C. Training Objectives

Our training paradigm formulates the task as a fully supervised, end-to-end regression problem, mapping sparse onset inputs directly to dense, expressive piano rolls containing velocity and duration information. While this setup shares structural similarities with image-to-image translation tasks, musical data presents unique challenges. Unlike natural images, where minor pixel shifts are often perceptually negli-

gible, in the musical domain, frame-level duration discrepancies can lead to significant rhythmic deviations. Furthermore, regression-based generation is highly susceptible to “sticking note” artifacts, where consecutive repeated notes are erroneously merged into a single sustained event due to the blurring of onset boundaries.

To address these issues and mitigate the convergence difficulties caused by data sparsity, we propose a Region-Adaptive Weighted Loss (RAW Loss). This objective function dynamically assigns importance weights to different regions of the piano roll—specifically prioritizing note onsets and active regions—to enforce structural precision. The total loss $\mathcal{L}_{\text{total}}$ is defined as:

$$\mathcal{L}_{\text{total}} = \frac{1}{|\Omega|} \sum_{i \in \Omega} \mathbf{W}_i \cdot \left(\lambda_1 |y_i - \hat{y}_i| + \lambda_2 (y_i - \hat{y}_i)^2 \right) \quad (2)$$

where y and $\hat{y}$ represent the ground truth and predicted values, respectively, and Ω denotes the set of all elements in the tensor. The point-wise loss is a weighted combination of ℓ_1 and ℓ_2 norms, controlled by $\lambda_1 = 1$ and $\lambda_2 = 4$, ensuring both numerical stability and gradient sharpness. The spatial weight tensor $\mathbf{W}$ is constructed to prioritize critical musical events. To handle overlapping regions of interest, we employ an element-wise maximum operation:

$$\mathbf{W} = \max(\alpha \cdot \mathbf{M}_{\text{fp}}, \beta \cdot \mathbf{M}_{\text{fg}}, \gamma \cdot \mathbf{M}_{\text{on}}) \quad (3)$$

Here, the masks are defined as indicator functions $\mathbb{I}(\cdot)$ tailored to specific error types:

- 1) **Foreground Mask** ($\mathbf{M}_{\text{fg}}$): $\mathbb{I}(y > 0)$. This focuses the model on correctly reconstructing active note events.
- 2) **False Positive Mask** ($\mathbf{M}_{\text{fp}}$): $\mathbb{I}(y = 0 \wedge \hat{y} > \epsilon)$, with $\epsilon = 10^{-2}$. This heavily penalizes “hallucinations” in silent regions to maintain sparsity.
- 3) **Onset Vicinity Mask** ($\mathbf{M}_{\text{on}}$): $\mathbb{I}(x_t > 0 \vee x_{t+1} > 0)$. Derived from the input onset tensor x , this mask covers the exact onset frame and its immediate temporal predecessor. This dilation is critical for enforcing sharp boundaries between repeated notes, thereby resolving the sticking note problem.

Based on empirical analysis of the loss landscape and hyperparameter search, we set the importance weights to $\alpha = 10$, $\beta = 30$, and $\gamma = 50$. This heavy weighting on onsets (γ) effectively counteracts the vanishing gradients often observed at note boundaries. Experimental monitoring confirms that this composite loss strategy significantly reduces sticking note artifacts and facilitates smooth, monotonic convergence of the training loss.

IV. EXPERIMENTS

A. Dataset

We constructed a large-scale multi-track dataset by aggregating XMIDI [18], GIGAMIDI [19], Lakh MIDI Dataset (LMD) [20], and Los Angeles MIDI Dataset (LAMMD) [21], resulting in a collection of over two million MIDI files. We

filtered this collection to retain samples containing a specific instrumentation quartet: *Acoustic Grand Piano*, *Acoustic Guitar (steel)*, *Acoustic Bass*, and *Drums*. We further cleaned the data by removing approximately 4% of noisy samples that contained irregular performance patterns. The final dataset comprises **87,989** 60-second MIDI segments, totaling over 1,466 hours of multi-track MIDI data. We partitioned the dataset into training, validation, and test sets (8:1:1), employing stratified sampling to maintain a consistent distribution of data sources across all splits.

B. Metrics

We adopt both objective and subjective metrics to evaluate the precision and expressiveness of the generated performance.

1) *Objective Evaluation*: We calculate metrics at both the frame level (for continuous contour) and the note level (for event reconstruction). We adopt the same note level metrics configurations as in existing AMT research [22], [23].

- **Duration F1**: Calculated as a frame-wise binary classification task (note on/off) with a time resolution of 10 ms.
- **Velocity MSE/MAE**: The Mean Squared Error and Mean Absolute Error between the predicted and ground truth velocities, calculated only on ground-truth note-on frames.
- **Velocity PCC** (ρ_{vel}): The Pearson Correlation Coefficient, used to assess the similarity of the velocity distribution and dynamic curves.
- **Note w/ Offset F1**: A predicted note is considered correct if its offset matches the reference within a tolerance of ± 50 ms or $\pm 20\%$ of the reference duration (whichever is larger).
- **Note w/ Vel F1**: A predicted note is considered correct if its velocity is within ± 12 (equivalent to 0.1 in normalized $[0, 1]$ space) of the reference.
- **Note w/ Offset & Vel F1**: A predicted note is considered correct if it matches both the offset criterion and the velocity criterion above.

For brevity, we report the aggregated scores across all instruments in our comparative analysis.

2) *Subjective Evaluation*: We conducted a MUSHRA-like listening test with 15 participants, all of whom possess an advanced level of musical proficiency. We selected four 30-second MIDI samples from the test set and synthesized the corresponding audio clips via *fluidsynth* [24]. For each sample, the subjects evaluated four versions: the **Reference** (ground truth), an **Anchor** (fixed velocity and duration), a **Baseline**, and the **Proposed**. Participants were required to rate the samples on a scale of 1 to 5 based on three criteria: *Rhythmic Expressiveness*, *Dynamic Expressiveness*, and *Overall Quality*. The Mean Opinion Score (MOS) is used to assess these criteria.

C. Experimental Results

1) *Objective Metrics Comparison*: We trained the model for 200 epochs on a single NVIDIA A100 GPU with a

TABLE I
EXPERIMENTAL RESULTS AND ABLATION STUDIES OF EACH MODULE COMPONENT ON NOTE- AND FRAME-LEVEL METRICS.

Model	Module			Note-level			Frame-level			
	C-T Patching	Skip Conv	RAW Loss	Off. F1 ↑	Vel. F1 ↑	O&V F1 ↑	Dur. F1 ↑	Vel-MSE ↓	Vel-MAE ↓	Vel- ρ ↑
Baseline				54.80%	54.94%	31.44%	81.40%	576.35	14.53	0.45
w/o C-T Patching		✓	✓	85.36%	71.29%	62.67%	93.57%	335.49	9.92	0.64
w/o RAW Loss	✓	✓		64.99%	60.35%	41.97%	84.23%	255.19	8.68	0.65
w/o Skip Conv	✓		✓	71.65%	59.18%	44.21%	87.08%	371.37	11.31	0.56
Proposed (CUT)	✓	✓	✓	86.09%	76.25%	68.04%	93.47%	190.47	6.96	0.71

TABLE II
PER-INSTRUMENT NOTE- AND FRAME-LEVEL PERFORMANCE ON THE TEST SET.

Instrument	Note-level			Frame-level			
	Note w/ Offset F1 ↑	Note w/ Vel F1 ↑	Note w/ Off.&Vel F1 ↑	Dur. F1 ↑	Vel-MSE ↓	Vel-MAE ↓	Vel- ρ ↑
Piano	83.32%	73.71%	63.94%	93.56%	156.68	7.51	0.80
Guitar	77.60%	72.57%	59.41%	92.97%	177.38	8.46	0.49
Bass	88.66%	80.42%	73.02%	95.01%	115.17	5.46	0.81
Drum	94.77%	78.29%	75.77%	92.32%	312.65	6.41	0.59
Average	86.09%	76.25%	68.04%	93.47%	190.47	6.96	0.71

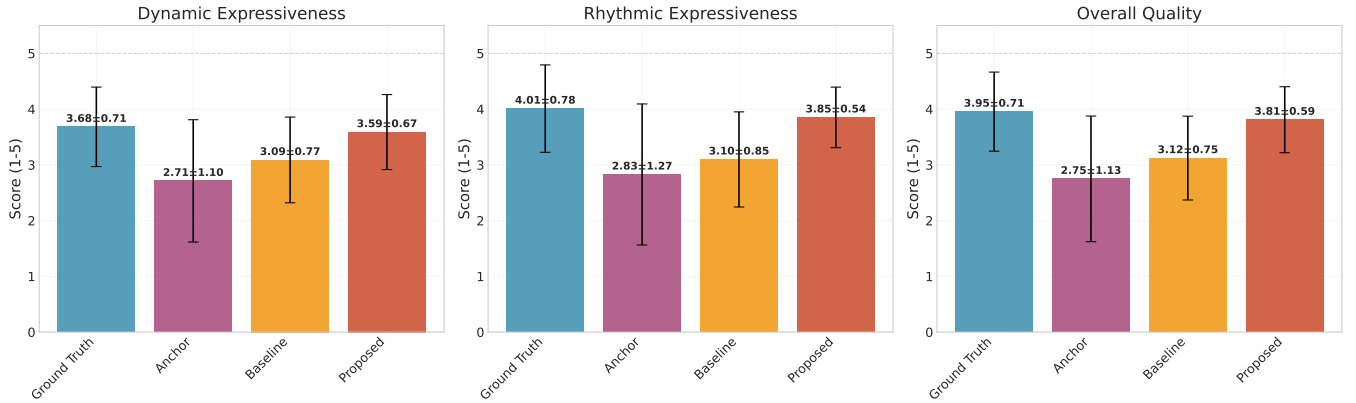


Fig. 3. Comparison of subjective MOS results on Dynamics Expressiveness, Rhythmic Expressiveness, and Overall Quality.

batch size of 32 and a learning rate of 8×10^{-5} , requiring approximately 133 hours of total training time. Given the absence of existing research on similar multi-track parameter estimation tasks, we implemented a basic Transformer baseline and several ablated versions for comparison. For the ablation study, we evaluated the contribution of three specific modules: the chromatic-temporal patching strategy (C-T Patching), the skip convolutional connection (Skip Conv), and the Region-Adaptive Weighted Loss (RAW Loss). As shown in Table I, the proposed model demonstrates significant performance improvements across both note-level and frame-level metrics. Notably, the skip convolution and weighted loss modules play a critical role in capturing local fine-grained details and delineating note boundaries. Although their corresponding ablated versions exhibit only minor discrepancies in frame-level metrics, they suffer a performance degradation of over 20 points in Note w/ Vel F1. Such degradation inevitably leads to a severe decline in the perceptual quality of the

synthesized audio. Furthermore, regarding the ablated version lacking temporal patch input, while its offset and duration detection performance remains comparable, its velocity estimation capability degrades significantly. This corroborates the vital importance of temporal context for accurate velocity estimation.

2) *Per-Instrument Performance*: We also present the performance of CUT across individual instrument tracks, as detailed in Table II. The results indicate that CUT maintains generally consistent performance across instruments, although certain anomalous patterns are observed. We attribute the relatively lower overall detection performance for piano and guitar to the diversity of their playing techniques and functional roles (serving simultaneously as melody and accompaniment), whereas bass and drums typically function in more singular roles. Regarding the unusually high Velocity MSE observed for drums, we assume that this stems from a small number of outliers, such as intensely dynamic drum solo segments. Excluding these anomalies, and despite the lack of directly

comparable methods, CUT demonstrates highly promising capabilities; for instance, the Velocity MAE on piano tracks is significantly lower than that of many existing piano velocity estimation models.

3) *Subjective Test Results:* We aggregated the Mean Opinion Scores (MOS) from the subjective evaluation and visualized them in Fig. 3. Due to subjective bias in individual rating preferences, the error bars (standard deviation) are relatively large, particularly for the Anchor. However, the rating patterns remain highly consistent: the Anchor received the lowest scores across all criteria, falling far below the other methods. The proposed model performs comparably to the Ground Truth, with MOS differences of only -0.09 , -0.16 , and -0.14 for Dynamic Expressiveness, Rhythmic Expressiveness, and Overall Quality, respectively. Furthermore, it achieved the lowest average standard deviation (0.60), highlighting the model’s superior capability in expression modeling and its consistency in user evaluation. In contrast, the Baseline showed only marginal improvements of $+0.38$, $+0.27$, and $+0.37$ over the Anchor, while lagging behind the proposed model by -0.50 , -0.75 , and -0.69 . This performance disparity underscores the challenging nature of this task and the effectiveness of our proposed modules. Based on these subjective and objective evaluations, we conclude that the proposed model is capable of predicting offset and velocity parameters for deadpan MIDI that closely approximate the expressiveness of real performances.

V. CONCLUSION

In this work, we formalized the task of Multi-track Expressive Parameter Estimation and presented CUT, a novel Chromatic-temporal U-Transformer designed to breathe life into deadpan, quantized MIDI sequences. By treating the problem as an image-to-image translation task, we demonstrated that high-fidelity performance modeling can be achieved without the need for explicit score features. Our empirical studies indicate that the proposed method is efficient for the low-informed performance rendering scenarios.

These findings have immediate practical implications for music production, offering a powerful tool for large-scale MIDI restoration, intelligent DAW assistance, and post-processing for Automatic Music Transcription. Future work will focus on the second stage of our proposed hierarchy—*Nuance Rendering*—to extend our framework from symbolic parameter estimation to direct, high-fidelity audio synthesis, further closing the loop on fully autonomous expressive music generation. We will also introduce more training data to our current model to support MIDI with more instrument tracks.

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